CBSE Class XII Board Examination 2027 — Expected Paper

BIOLOGY (Theory) · Code 044

Time allowed: 3 hours **Maximum marks: 70**

General instructions

1. This paper has 33 questions in five sections, A–E. All questions are compulsory.
2. Section A: Q1–16 are MCQs of 1 mark each (Q13–16 are Assertion–Reason). Section B: Q17–21, 2 marks each. Section C: Q22–28, 3 marks each.
3. Section D: Q29–30 are case-based questions of 4 marks each. Section E: Q31–33, 5 marks each.
4. This is an original practice paper built on the latest CBSE sample-paper pattern. It is not an official CBSE paper.

Section A — Multiple choice (1 mark each). Questions 13–16 are Assertion–Reason.

1. Which of these is haploid? **[1]**

(a) Nucellus
(b) Microspore mother cell
(c) Pollen grain
(d) Integument

2. The filiform apparatus is present in the **[1]**

(a) egg cell
(b) synergids
(c) antipodals
(d) central cell

3. Ovulation is triggered by a surge of **[1]**

(a) FSH
(b) LH
(c) progesterone
(d) oxytocin

4. The number of chromosomes in a human primary spermatocyte is **[1]**

(a) 23
(b) 46
(c) 92
(d) 69

5. Which is a barrier method of contraception? **[1]**

(a) Copper-T
(b) Diaphragm
(c) Oral pills
(d) Vasectomy

6. Haemophilia is a **[1]**

(a) sex-linked recessive disorder
(b) autosomal dominant disorder
(c) chromosomal disorder
(d) Y-linked disorder

7. Okazaki fragments are synthesised on the **[1]**

(a) leading strand
(b) lagging strand
(c) mRNA
(d) both strands equally

8. In the lac operon, the repressor protein is coded by the **[1]**

(a) z gene
(b) y gene
(c) a gene
(d) i gene

9. An example of analogous organs is **[1]**

(a) forelimbs of whale and bat
(b) wings of butterfly and bird
(c) thorns of Bougainvillea and tendrils of Cucurbita
(d) heart of vertebrates

10. Passive immunity is provided by **[1]**

(a) the polio vaccine
(b) antitoxins/colostrum
(c) the BCG vaccine
(d) a natural infection

11. In EcoRI, the letter 'E' stands for **[1]**

(a) Enzyme
(b) Escherichia
(c) Endonuclease
(d) Eco

12. The Bt toxin is coded by **[1]**

(a) cry genes
(b) nif genes
(c) lac genes
(d) hox genes

13. **Assertion (A):** The endosperm of angiosperms is triploid. **[1]**
Reason (R): It forms by the fusion of two polar nuclei with a male gamete.

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** The F_2 genotypic ratio of a monohybrid cross is 1 : 2 : 1. **[1]**
Reason (R): The two alleles of a gene segregate during gamete formation.

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** Decomposition is faster in warm and moist conditions. **[1]**
Reason (R): Decomposer micro-organisms are more active at higher temperature and moisture.

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

16. **Assertion (A):** Cancer cells show contact inhibition. **[1]**
Reason (R): Normal cells stop dividing when they contact neighbouring cells.

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

Section B — Very short answer (2 marks each)

17. Differentiate between geitonogamy and xenogamy. **[2]**

18. How does a copper-releasing IUD prevent pregnancy? Give one advantage of IUDs. **[2]**

19. Give two reasons why DNA is a better genetic material than RNA. **[2]**

20. Name the causative agent of typhoid and the test used to confirm it. Write one symptom. **[2]**

21. What are biofertilisers? Give two examples. **[2]**

Section C — Short answer (3 marks each)

22. Describe the structure of a mature embryo sac. What is double fertilisation? **[3]**

23. Describe the phases of the menstrual cycle and the role of hormones in each. **[3]**

24. State the law of independent assortment. Explain it with a dihybrid cross between round yellow (RRYY) and wrinkled green (rryy) pea plants. **[3]**

25. Describe the three steps of transcription in bacteria. **[3]**

26. State the Hardy–Weinberg principle. List the five factors that affect genetic equilibrium. **[3]**

27. Describe the three steps of PCR. **[3]**

28. Explain with examples (a) mutualism, (b) competition and (c) brood parasitism. **[3]**

Section D — Case-based questions (4 marks each)

29. **Case study: HIV and AIDS.** HIV is a retrovirus spread by sexual contact, infected blood, shared needles and from mother to foetus. It infects helper T-cells and slowly weakens immunity. **[4]**
(i) Which cells does HIV mainly attack?
(ii) Name the test used to diagnose AIDS.
(iii) Why is HIV called a retrovirus?
(iv) Why do AIDS patients suffer from opportunistic infections?

30. **Case study: ecological pyramids.** The number, biomass or energy at successive trophic levels can be drawn as pyramids. By Lindeman's 10% law, only about 10% of energy passes to the next level. **[4]**
(i) Give an example of an inverted pyramid of biomass.
(ii) Why is the pyramid of energy always upright?
(iii) If producers fix 10,000 J, how much energy reaches the tertiary consumers?
(iv) What is standing crop?

Section E — Long answer (5 marks each)

31. Describe spermatogenesis and oogenesis with labelled flow charts. Give two differences between them and explain their hormonal control. **[5]**

32. Explain the lac operon. How is it switched off in the absence of lactose and switched on in its presence? Why is it called negative regulation? **[5]**

33. What is meant by the 'Evil Quartet'? Explain each cause of biodiversity loss. Differentiate between in-situ and ex-situ conservation with two examples each. **[5]**

Marking scheme — Biology (044)

Value points and mark split

Section A — Multiple choice (1 mark each). Questions 13–16 are Assertion–Reason.

1. (c) Pollen grains form by meiosis of the MMC.

2. (b) Synergids; it guides the pollen tube.

3. (b) The mid-cycle LH surge.

4. (b) 46; it is diploid before meiosis I.

5. (b) Diaphragm.

6. (a) X-linked recessive.

7. (b) Lagging strand.

8. (d) The regulator i gene.

9. (b) Similar function, different structure (convergent evolution).

10. (b) Ready-made antibodies are given.

11. (b) The genus Escherichia.

12. (a) cry genes of Bacillus thuringiensis.

13. (a) Triple fusion: 2n + n = 3n.

14. (a) Segregation gives the TT : Tt : tt ratio.

15. (a) Low temperature and anaerobiosis slow decomposition.

16. (d) Cancer cells have lost contact inhibition, so A is false.

Section B — Very short answer (2 marks each)

17. Geitonogamy: pollen goes to the stigma of another flower on the same plant (genetically like selfing; needs an agent). Xenogamy: pollen goes to the stigma of a flower on a different plant, bringing genetically different pollen (true cross-pollination). [2]

18. Cu ions suppress sperm motility and fertilising capacity and increase phagocytosis of sperm in the uterus. Advantage: long-acting, reversible and ideal for spacing children. [1½ + ½]

19. (i) DNA is chemically more stable: no 2′-OH, and it is double-stranded. (ii) It has thymine instead of uracil, so mutations are fewer and repair is possible; RNA mutates faster. [1 + 1]

20. Salmonella typhi; the Widal test. Symptoms: sustained high fever (39–40 °C), weakness, stomach pain, loss of appetite. [½ + ½ + 1]

21. Organisms that enrich the nutrient quality of the soil. Examples: Rhizobium (symbiotic N_2 fixation), Azotobacter, Anabaena or Nostoc (cyanobacteria), mycorrhiza (Glomus). [1 + 1]

Section C — Short answer (3 marks each)

22. 7-celled, 8-nucleate: the egg apparatus (egg cell + 2 synergids with filiform apparatus) at the micropylar end, 3 antipodals at the chalazal end, and a central cell with 2 polar nuclei. Double fertilisation: one male gamete + egg → zygote (2n) (syngamy); the other male gamete + 2 polar nuclei → PEN (3n) (triple fusion). [2 + 1]

23. Menstrual phase (days 1–5): the endometrium breaks down when progesterone falls. Follicular phase (6–13): FSH and LH drive follicle growth; oestrogen rebuilds the endometrium. Ovulation (about day 14): LH surge. Luteal phase (15–28): the corpus luteum secretes progesterone to maintain the endometrium; without fertilisation it degenerates. [3]

24. When two pairs of traits are combined, the segregation of one pair is independent of the other. F_1: RrYy (round yellow). F_1 gametes: RY, Ry, rY, ry in equal proportion. F_2: 9 round yellow : 3 round green : 3 wrinkled yellow : 1 wrinkled green. [1 + 2]

25. Initiation: RNA polymerase with the σ factor binds the promoter. Elongation: the polymerase adds ribonucleotides 5′→3′ using the template strand. Termination: the ρ factor helps release the RNA at the terminator. [1 × 3]

26. Allele and genotype frequencies in a population stay constant across generations: $p^2 + 2pq + q^2 = 1$. Factors: gene migration (gene flow), genetic drift, mutation, genetic recombination, natural selection. [1 + 2]

27. Denaturation: heating to about 94 °C separates the DNA strands. Annealing: primers bind the complementary sequences at about 50–60 °C. Extension: Taq polymerase extends the primers at about 72 °C. Repeated cycles amplify the DNA exponentially. [1 × 3]

28. (a) Both partners benefit: lichens (fungus + alga), mycorrhiza, fig–wasp. (b) Both are harmed: Gause's competitive exclusion; Abingdon tortoise vs goats in the Galapagos. (c) The cuckoo lays eggs in the crow's nest and the host raises its young. [1 × 3]

Section D — Case-based questions (4 marks each)

29. (i) Helper T-lymphocytes ($CD4^+$), and macrophages. (ii) ELISA. (iii) It has an RNA genome copied into DNA by reverse transcriptase. (iv) Falling helper T-cell numbers cripple immunity, so normally harmless microbes such as Mycobacterium and Toxoplasma cause disease. [1 × 4]

30. (i) In the sea, phytoplankton biomass is less than zooplankton/fish biomass. (ii) Energy is lost as heat at every level (second law of thermodynamics). (iii) 10,000 → 1,000 → 100 → 10 J. (iv) The mass of living material at a trophic level at a given time. [1 × 4]

Section E — Long answer (5 marks each)

31. Spermatogenesis: spermatogonia (2n) → mitosis → primary spermatocytes (2n) → meiosis I → secondary spermatocytes (n) → meiosis II → spermatids → spermiogenesis → sperm (4 per primary spermatocyte). Control: GnRH → LH (Leydig cells, androgens) and FSH (Sertoli cells). Oogenesis: oogonia (fetal life) → primary oocytes arrested in prophase I → at puberty, meiosis I → secondary oocyte + 1st polar body; meiosis II completes only on sperm entry → ovum + 2nd polar body. Differences: 4 gametes vs 1 functional ovum; continuous from puberty vs starts before birth and is cyclic. [5]

32. Components: regulator gene i (makes the repressor), promoter, operator, and structural genes z (β-galactosidase), y (permease), a (transacetylase). No lactose: the repressor binds the operator, blocking RNA polymerase, so there is no transcription. Lactose present: allolactose (inducer) binds the repressor and inactivates it, so RNA polymerase transcribes z, y and a. Negative regulation: the repressor switches the operon off. [5]

33. Evil Quartet: (1) habitat loss and fragmentation (e.g. clearing of rainforests); (2) over-exploitation (Steller's sea cow, passenger pigeon); (3) alien species invasion (Nile perch in Lake Victoria, Parthenium, Lantana, water hyacinth); (4) co-extinctions (host–parasite, plant–pollinator). In-situ protects species in their natural habitat: national parks, wildlife sanctuaries, biosphere reserves, sacred groves, hotspots. Ex-situ protects them outside it: zoos, botanical gardens, seed banks, cryopreservation, tissue culture. [5]